

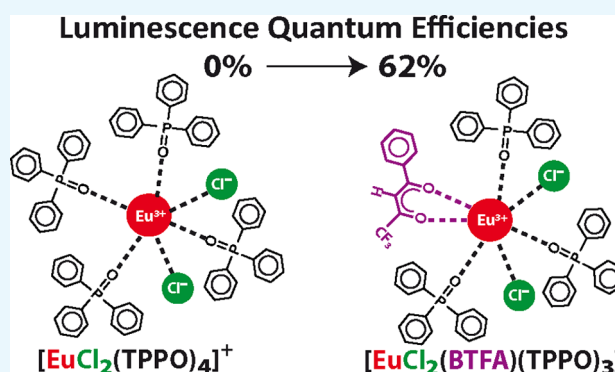
Europium Complexes: Luminescence Boost by a Single Efficient Antenna Ligand

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Supporting Information

ABSTRACT: We advance the concept that a single efficient antenna ligand substituted in or added to an otherwise weakly luminescent europium complex is enough to significantly boost its luminescence. Our results, on the basis of photophysical measurements on 5 novel europium complexes and 15 known ones, point in the direction that ligand dissimilarity and ligand diversity are all concepts that clearly play a fundamental role in the luminescence of europium complexes. We show that it is important that a symmetry breaker ligand exists in the complex to enhance ligand dissimilarity and ligand diversity, all mainly affecting the nonradiative decay rate by reducing it. Because the presence of at least one antenna ligand is also obviously necessary, the optimal and the most cost-effective situation can be achieved by adding a single coordination symmetry breaker that is also an efficient antenna, such as 1-(2-thenoyl)-3,3,3-trifluoroacetone or 4,4,4-trifluoro-1-phenyl-1,3-butanedione. In such cases the quantum efficiency, η , is decidedly boosted, as can be verified by going from complex $[\text{EuCl}_2(\text{TPPO})_4]^+$ with $\eta = 0\%$ to the novel complex $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$, where TPPO stands for triphenylphosphine oxide, with $\eta = 62\%$.



1. INTRODUCTION

Luminescence in europium complexes occurs through the so-called antenna effect:¹ when the complexes are illuminated with ultraviolet (UV) light, the UV photons are absorbed by the ligands and the energy is then transferred to the europium ion, which subsequently decays by emitting visible light in the red-orange region.

Luminescent europium complexes can be utilized in several technologies and biological applications, such as light-emitting diode,^{2–4} organic light-emitting diode,^{5,6} bioimaging,^{7,8} sensing and targeting specific DNA structures,⁹ melamine detection in milk products,¹⁰ and cellular imaging.¹¹

Hence, due to the innumerable applications, intensification of the brightness of luminescence by chemically modifying the europium complexes is an active area of research.

The quest to increase brightness requires designing a complex capable of (i) being a strong absorbent of incident UV light, (ii) transferring the absorbed energy to the europium ion, and (iii) decaying mostly via radiative channels. Of course, the decay via radiative channels will be preferred if the nonradiative decay rate is minimized. Controlling all these phenomena synchronously is, of course, a challenging scientific pursuit.

The quantum yield of luminescence of a europium complex ϕ , also called the overall quantum yield, is equal to the number of emitted photons divided by the number of absorbed photons.¹² The quantum efficiency η , also called intrinsic quantum yield, $Q_{\text{Eu}}^{\text{Eu}}$ is the probability of europium emission

once the europium is excited, and is equal to $A_{\text{rad}}/(A_{\text{rad}} + A_{\text{nrad}})$, where A_{rad} and A_{nrad} are, respectively, the radiative and the nonradiative decay rates of the excited states of the europium ion, most notably the $^5\text{D}_0$ one.¹² The inverse of the emission lifetime τ is equal to $A_{\text{rad}} + A_{\text{nrad}}$.¹²

Recently, our research group reported a comprehensive strategy to boost luminescence in europium complexes by mixing ligands and exemplified this strategy by mixing nonionic ligands of the type phosphine oxide and sulfoxide compounds on ternary europium complexes.¹³ Results showed that increasing the coordination diversity by the presence of mixed nonionic ligands caused a boost, both in the emission quantum yield, ϕ , as well as on the emission quantum efficiency, η .¹³ Later, it was also found out that mixing ligands also caused a boost in the radiative decay rates, A_{rad} .¹⁴

Europium β -diketonate complexes with repeating ligands are a well-known and highly researched class of luminescent complexes.^{15–18} However, europium β -diketonate complexes with mixed ligands are not routine compounds. Actually they are mostly novel complexes that are only now being researched for their luminescent potential^{13,12,19} in light of our boost strategy mentioned above.¹³

In this article, we seek to clarify the finer details and aspects influencing the luminescence of europium complexes. Accord-

Received: August 14, 2017

Accepted: September 18, 2017

Published: October 17, 2017

Table 1. Luminescence Data for All 20 Complexes Considered^{a,b}

class	complex	τ (ms)	A_{tot} (s ⁻¹)	τ_{rad} (ms)	A_{rad} (s ⁻¹)	τ_{nrad} (ms)	A_{nrad} (s ⁻¹)	η (%)
(i)	[EuCl ₂ (TPPO) ₄]Cl·3H ₂ O ²⁰	----	----	----	----	----	----	~0
(ii)	[EuCl ₂ (DBM)(TPPO) ₃] ²¹	----	----	----	----	----	----	~0
	[EuCl ₂ (TTA)(TPPO) ₃] ²¹	0.465	2151	0.943	1061	0.917	1090	49
	[EuCl ₂ (BTFA)(TPPO) ₃]	0.583	1715	0.941	1063	1.531	652	62
(iii)	[EuCl(DBM) ₂ (TPPO) ₂]	0.131	7634	0.779	1283	0.157	6351	17
	[EuCl(TTA) ₂ (TPPO) ₂]	0.425	2352	1.229	814	0.650	1539	35
	[EuCl(BTFA) ₂ (TPPO) ₂]	0.503	1987	0.876	1142	1.183	845	57
(iv)	[EuCl(DBM)(TTA)(TPPO) ₂]	0.393	2545	1.193	838	0.586	1707	33
	[EuCl(DBM)(BTFA)(TPPO) ₂] ²¹	0.420	2380	1.241	806	0.635	1574	34
	[EuCl(TTA)(BTFA)(TPPO) ₂] ²¹	0.423	2364	1.215	823	0.649	1541	35
(v)	[Eu(DBM) ₃ (TPPO) ₂] ¹⁴	----	----	2.985	335	----	----	~0
	[Eu(TTA) ₃ (TPPO) ₂] ¹³	0.350	2857	1.256	796	0.485	2061	28
	[Eu(BTFA) ₃ (TPPO) ₂] ¹³	0.367	2725	1.088	919	0.554	1806	34
(vi)	[Eu(DBM) ₂ (TTA)(TPPO) ₂] ¹⁹	0.415	2410	0.924	1082	0.753	1328	45
	[Eu(TTA) ₂ (DBM)(TPPO) ₂] ¹⁹	0.424	2359	0.951	1052	0.765	1307	45
	[Eu(DBM) ₂ (BTFA)(TPPO) ₂] ¹⁹	0.423	2364	0.985	1015	0.741	1349	43
	[Eu(BTFA) ₂ (DBM)(TPPO) ₂] ¹⁹	0.473	2114	0.884	1131	1.017	983	53
	[Eu(TTA) ₂ (BTFA)(TPPO) ₂] ¹⁹	0.435	2299	0.915	1093	0.829	1206	48
	[Eu(BTFA) ₂ (TTA)(TPPO) ₂] ¹⁹	0.458	2184	0.945	1058	0.888	1126	48
(vii)	[Eu(DBM)(TTA)(BTFA)(TPPO) ₂] ¹⁹	0.434	2303	0.967	1034	0.788	1269	45

^aLifetimes, τ ; total decay rates, A_{tot} ; radiative lifetimes, τ_{rad} ; radiative decay rates, A_{rad} ; nonradiative lifetimes, τ_{nrad} ; nonradiative decay rates, A_{nrad} ; and quantum efficiency, η . ^bA sequence of dashes, ----, indicates that data could not be measured due to very poor luminescence.

ingly, we studied a set of 20 different europium complexes to unveil and make explicit the structural aspects most determining of their photophysical properties, such as ligand diversity, ligand dissimilarity, and centrosymmetry breaking. The compounds were selected in a combinatorial manner so that the various structural effects on luminescence could be isolated. For this purpose, we synthesized five novel chloride complexes: [EuCl₂(BTFA)(TPPO)₃]; [EuCl(TTA)₂(TPPO)₂]; [EuCl(BTFA)₂(TPPO)₂]; [EuCl(DBM)₂(TPPO)₂]; and [EuCl(DBM)(TTA)(TPPO)₂]. We also measured in this article the luminescence properties of five other chloride complexes that had syntheses and characterizations recently reported by our research group: [EuCl₂(TPPO)₄]Cl·3H₂O,²⁰ [EuCl₂(TTA)(TPPO)₃],²¹ [EuCl₂(DBM)(TPPO)₃],²¹ [EuCl₂(TTA)(BTFA)(TPPO)₃],²¹ and [EuCl₂(DBM)(BTFA)(TPPO)₃].²¹ Finally, to complete the luminescence databank for the combinatorial set of compounds, we included the previously determined luminescence properties of ternary europium(III) complexes of the types [Eu(β)₃(TPPO)₂],^{14,22,23} [Eu(β)₂(β')(TPPO)₂],¹⁹ and [Eu(β)(β')(β'')(TPPO)₂],¹⁹ where $\beta \neq \beta' \neq \beta''$ can be either 1-(2-thenoyl)-3,3,3-trifluoroacetone (TTA), 4,4,4-trifluoro-1-phenyl-1,3-butanedione (BTFA), or 1,3-diphenylpropane-1,3-dione (DBM). We then examined the effect of each type of ligand coordinated to the europium ion on the radiative decay rates using the chemical partition of A_{rad} .²⁴ We further evaluated the effect of ligand dissimilarity on the nonradiative decay rates. We conclude by advancing the concept that a single ligand dissimilarity and ligand diversity enhancer antenna of high efficiency substituted in or added to an otherwise weakly luminescent europium complex is enough to significantly boost its luminescence.

2. RESULTS AND DISCUSSION

Table 1 presents luminescence data for the 5 novel europium complexes and the 15 known ones, belonging to the seven classes: (i) [EuCl₂(TPPO)₄]Cl·3H₂O; (ii)

[EuCl₂(β)(TPPO)₃]; (iii) [EuCl(β)₂(TPPO)₂]; (iv) [EuCl(β)(β')(TPPO)₂]; (v) [Eu(β)₃(TPPO)₂]; (vi) [Eu(β)₂(β')(TPPO)₂]; and (vii) [Eu(β)(β')(β'')(TPPO)₂], where $\beta \neq \beta' \neq \beta''$ in all types stands for ionic β -diketonate ligands, which function as antennae and can be either TTA, BTFA, or DBM.

As such, we vary the ionic ligands in a combinatorial manner and maintain the same nonionic ligands throughout. We chose TPPO as the nonionic ligand because it is bulky, relatively rigid and therefore, in principle, should not contribute significantly to the nonradiative processes. The novel chloride europium complexes, [EuCl₂(BTFA)(TPPO)₃], [EuCl(TTA)₂(TPPO)₂], [EuCl(BTFA)₂(TPPO)₂], [EuCl(DBM)₂(TPPO)₂], and [EuCl(DBM)(TTA)(TPPO)₂], were all synthesized and had their photophysical properties measured, as mentioned in this article. The other chloride complexes, [EuCl₂(TPPO)₄]Cl·3H₂O, [EuCl₂(TTA)(TPPO)₃], [EuCl₂(DBM)(TPPO)₃], [EuCl₂(TTA)(BTFA)(TPPO)₃], and [EuCl₂(DBM)(BTFA)(TPPO)₃], had been previously synthesized^{20,21} and had their photophysical properties measured in this work (as can be seen in the [Supporting Information](#)). The luminescence data for complexes of classes (v)–(vii) were obtained from previous works by our research group.^{13,14,19}

Complex [EuCl₂(TPPO)₄]Cl·3H₂O presents a luminescence which is so low that its quantum efficiency η can be taken as being essentially 0%. This can be explained by the absence of efficient antennae in the complex structure, which contains only chlorides and TPPOs as ligands.

Of the β -diketonate ligands, only TTA and BTFA display high efficiencies as antennae. Indeed, the efficiency of DBM, as an antenna, is very poor, as we verified in the case of the complex [EuCl₂(DBM)(TPPO)₃], which we prepared by adding a DBM ligand to [EuCl₂(TPPO)₄]Cl·3H₂O and by eliminating its chloride counter ion. This complex displays such a low luminescence that is essentially immeasurable.

Clearly, for luminescence to emerge, it is important that at least one good antenna ligand be present in the structure. However, our results indicate that adding more than one antenna does not necessarily improve the quantum efficiency further, as can be seen in Table 1, when complexes with only one antenna ligand display the largest quantum efficiencies η .

Indeed, the fact that adding more of the same antenna ligand to a complex does not necessarily improve the quantum efficiency indicates that there are more factors at play, factors that deserve a closer and more detailed examination.

2.1. Coordination Centrosymmetry Breaking Ligands.

The Laporte rule²⁵ states that atomic f–f transitions are forbidden in centrosymmetric chemical environments. Therefore, to make radiative decay less forbidden and thus more likely to occur, relaxation of centrosymmetry must take place. This, in itself, is well-known.²⁶ Only recently, it was advanced and experimentally proven that a significant manner in which this can be obtained is to move from same ligand complexes to mixed-ligand ones.¹³ Indeed, by strategically mixing ligands, one can break centrosymmetry in a much more decisive manner than, for example, by thermal vibrations.

The reduction in quantum efficiency displayed in Table 1, when more antennae are added to the complex, can be understood in terms of which ligands make centrosymmetry happen and which ligands break the centrosymmetry.

Triphenylphosphine oxide (TPPO), with its three phenyl groups attached to a phosphine oxide, is a very bulky ligand, so when there are only two of them attached to the europium ion, they prefer to place themselves opposite to each other, thereby enhancing the centrosymmetry of the complex, as in Figure 1.

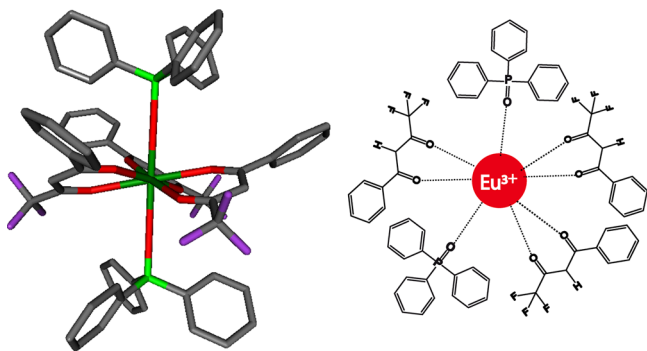


Figure 1. Fully optimized Sparkle/RM1 geometry and the chemical structure of the complex $[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$.

This preference for opposite placements with respect to the europium ion, predicted by Sparkle/RM1,²⁷ had already been established by crystallography for two of the complexes $[\text{Eu}(\text{TTA})_3(\text{TPPO})_2]$ ²² and $[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$.²³

Moreover, in complexes of the type $[\text{Eu}(\beta)_3(\text{TPPO})_2]$, the three β -diketonates tend to place themselves in a triangular positioning around the europium ion in the equatorial plane, as can also be seen in Figure 1.

On the other hand, when there are three TPPOs, Sparkle/RM1 predicts quite logically that two of these remain relatively opposite to each other, but the third is necessarily placed in another position without a counterpart. This removes some symmetry, which occurs when several identical β -diketonates are all coordinated to a single center. Therefore, centrosymmetry is necessarily reduced because TPPO is very different from the other ionic ligands, as can be seen in Figure 2. This

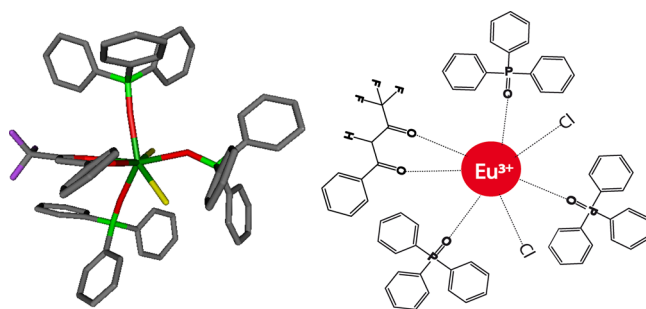


Figure 2. Fully optimized Sparkle/RM1 geometry and the chemical structure of the complex $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$.

explains why complexes with three TPPOs have larger quantum efficiencies when compared with complexes with two TPPOs, as shown in Table 1. Clearly, complexes with three TPPOs cannot be centrosymmetric and, provided the requirement that at least one antenna should be present in the complex is satisfied, luminescence ensues.

2.2. Chloride as a Ligand Dissimilarity Enhancer. In complexes of the kind $[\text{EuCl}(\beta)_2(\text{TPPO})_2]$, chloride now replaces one of the β -diketonates in the equatorial plane, as in Figure 3. Given the fact that it consists of a single atom, the

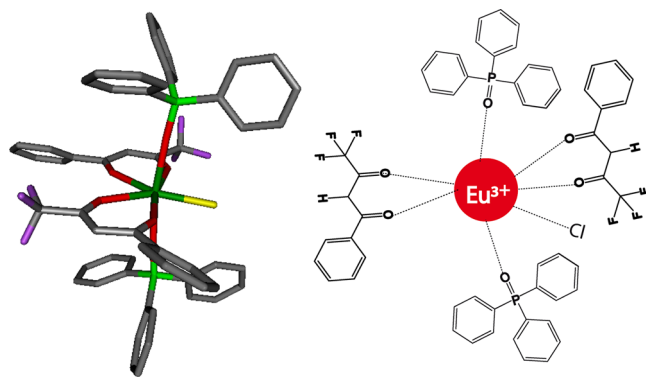


Figure 3. Fully optimized Sparkle/RM1 geometry and the chemical structure of the complex $[\text{EuCl}(\text{BTFA})_2(\text{TPPO})_2]$.

chloride dramatically increases ligand dissimilarity and makes the situation in the equatorial plane significantly less centrosymmetric, as can be seen in Figure 3.

Chloride, by being a single atom, turns out to be a very efficient ligand dissimilarity enhancer and coordination symmetry breaker, which can boost luminescence in a significant manner. Indeed, from Table 1, the average quantum efficiency of the complexes $[\text{Eu}(\beta)_3(\text{TPPO})_2]$, of $\eta = 31\%$ ¹³ (with β ligand being either TTA or BTFA) is boosted by 48% in complexes with a single chloride ligand $[\text{EuCl}(\beta)_2(\text{TPPO})_2]$ to $\eta = 46\%$.

As mentioned above, DBM, being a β -diketonate, is an antenna, albeit with a very low efficiency. We then decided to examine whether the replacement of a DBM by a chloride would also significantly boost the quantum efficiency of the luminescence of $[\text{Eu}(\text{DBM})_3(\text{TPPO})_2]$,¹⁴ which is immeasurably low. Indeed, that was the case, as can be verified by examining Table 1. The quantum efficiency jumped from essentially 0% for $[\text{Eu}(\text{DBM})_3(\text{TPPO})_2]$ ¹⁴ to 17% for $[\text{EuCl}(\text{DBM})_2(\text{TPPO})_2]$.

2.3. β -Diketonate as a Coordination Symmetry Breaker. If a simple single-atom ligand, as chloride, is capable of

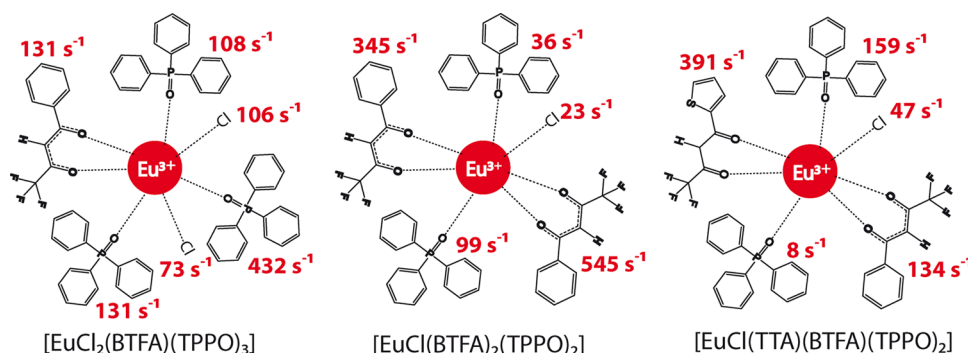


Figure 4. Chemical structures and chemical partition of the radiative decay rates $A_{\text{rad}'}$ per ligand for complexes $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$, $[\text{EuCl}(\text{BTFA})_2(\text{TPPO})_2]$, and $[\text{EuCl}(\text{TTA})(\text{BTFA})(\text{TPPO})_2]$.

breaking the coordination symmetry of europium complexes and boosting luminescence, then a different β -diketonate should play a similar role as a ligand dissimilarity enhancer and coordination symmetry breaker.

As such, let us consider the difference in luminescence in going from complexes $[\text{Eu}(\beta)_3(\text{TPPO})_2]$ to complexes $[\text{Eu}(\beta)_2(\beta')(\text{TPPO})_2]$, where β' behaves as a ligand dissimilarity enhancer and coordination symmetry breaker. Results are summarized in Table 1 for $\beta = \text{TTA}$ and BTFA .

Again, the average quantum efficiency of complexes of the type $[\text{Eu}(\beta)_3(\text{TPPO})_2]$ is $\eta = 31\%$, whereas by adding a different β -diketonate as a ligand dissimilarity enhancer and coordination symmetry breaker, $[\text{Eu}(\beta)_2(\beta')(\text{TPPO})_2]$, the quantum efficiency is boosted by 52% to an average of $\eta = 47\%$, quite similar to the boost of 48% obtained by employing chloride as a coordination symmetry breaker. The chemical nature of the coordination symmetry breaking ligand per se is thus seemingly not relevant. It is important, though, that a symmetry breaker ligand exists so that ligand dissimilarity and ligand diversity are enhanced.

2.4. Antennae as Ligand Dissimilarity Enhancers.

Because we established the need for at least one efficient antenna ligand and also one ligand dissimilarity enhancer and coordination symmetry breaker ligand, let us now examine the cases when the antenna and the ligand dissimilarity and coordination symmetry breaker roles are played out, simultaneously, by a single ligand. That could turn out to be perhaps an optimal condition.

Now, let us first consider complex $[\text{Eu}(\text{DBM})_3(\text{TPPO})_2]$, which does not contain any highly efficient antennae. By replacing a DBM with either a BTFA or a TTA, which behave as good antennae, as ligand dissimilarity enhancers and as coordination symmetry breakers, the boost in luminescence is dramatic, from essentially zero to an average of $\eta = 44\%$.

Likewise, consider now the complex $[\text{EuCl}_2(\text{TPPO})_4]\cdot\text{Cl}\cdot 3\text{H}_2\text{O}$, which also does not have any antennae in its formula and therefore displays a quantum efficiency of essentially zero. The addition of a DBM as a symmetry breaker, leading to $[\text{EuCl}_2(\text{DBM})(\text{TPPO})_3]$, does not help because DBM is not an efficient antenna. Indeed, the quantum efficiency remains immeasurably low. However, by adding a symmetry breaker, which is also a very effective antenna, such as TTA or BTFA, the quantum efficiency is boosted tremendously, to a maximum of $\eta = 62\%$ for $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$. The fact that this recordist complex has two chloride ions as ligands adds to the reality that the chemical nature of the other ligands that are neither antennae nor ligand dissimilarity enhancers or

coordination symmetry breakers plays only a supportive role in the luminescence phenomenon, not a protagonist one.

2.5. Several Ligand Dissimilarity Enhancers. If a single coordination symmetry breaker is capable of significantly boosting the quantum efficiency of luminescence of europium complexes, one could guess that perhaps several coordination symmetry breakers would lead to an even more luminescent complex. However, we verified that the presence of several symmetry breakers not only quickly exhausts their effect on the luminescence, but also starts to slowly interfere negatively, one with the other. From Table 1, the quantum efficiencies of the fully mixed ionic ligand complexes of the formulae $[\text{EuCl}(\beta)(\beta')(\text{TPPO})_2]$ have an average $\eta = 34\%$, a value which is equivalent to the average $\eta = 36\%$ of complexes $[\text{EuCl}(\beta)_2(\text{TPPO})_2]$. Likewise, the quantum efficiency of the fully mixed ionic ligand complex $[\text{Eu}(\text{DBM})(\text{BTFA})(\text{TTA})(\text{TPPO})_2]$ is $\eta = 45\%$,¹⁹ a value which is equivalent to the average $\eta = 47\%$ ¹⁹ of all six complexes of the type $[\text{Eu}(\beta)_2(\beta')(\text{TPPO})_2]$. Thus, the chemical difficulty involved in the synthesis of a fully mixed-ligand complex does not translate into further luminescence boost and is therefore not justified.

2.6. Chemical Partition of $A_{\text{rad}'}$. The radiative decay rate A_{rad} of a given complex is a more direct measure of how less forbidden the f–f transitions become in the presence of the ligands. Recently, our research group introduced the chemical partition of $A_{\text{rad}'}$,²⁴ where $A_{\text{rad}'}$ is the sum of the contributions of the $^5\text{D}_0 \rightarrow ^7\text{F}_J$ ($J = 2, 4$, and 6) transitions, which represent more than 90% of A_{rad} . Such a partition provides a very useful assessment, in chemical terms, of the role of each ligand as a facilitator of the radiative decay in the complex. The larger the value of A_{rad} and $A_{\text{rad}'}$, the more is the complex luminescence via radiative channels.

Accordingly, to calculate the chemical partition, we first fully optimized the geometries of the complexes with the Sparkle/RM1 model.²⁷ The quality of the obtained geometries was then assessed from the values of the Q , D , and C parameters,²⁸ as arrived at by the LUMPAC software²⁹ from the emission spectra of the europium complexes. LUMPAC adjusts these Q , D , and C parameters for each europium complex to reproduce the various experimentally obtained intensity parameters $\Omega_{\lambda, \text{exp}}$ ^{30,31} with $\lambda = 2$ and 4 . Table S1 of the Supporting Information shows all Q , D , and C parameters from LUMPAC, as well as the values of $\Omega_{\lambda, \text{exp}}$ with $\lambda = 2$ and 4 and the values of $\Omega_{\lambda, \text{theo}}$ with $\lambda = 2, 4$, and 6 .

The fitted parameters Q , D , and C must obey the acceptance criterion $D/C > 1$ for a unique adjustment.²⁸ If the ratio $D/C \leq$

Table 2. Values of A_{rad}' and the Average Values of Their Chemical Partitions Per Type of Ligand^a

complex	$A_{\text{rad}}' \text{ (s}^{-1}\text{)}$	average $A_{\text{rad}}^{\text{ionic}} \text{ (s}^{-1}\text{)}$		average $A_{\text{rad}}^{\text{nonionic}} \text{ (s}^{-1}\text{)}$
		Cl [−]	β -diketonate	
[EuCl ₂ (TTA)(TPPO) ₃] ²¹	974	29	179	246
[EuCl ₂ (BTFA)(TPPO) ₃]	981	90	131	224
[EuCl(DBM) ₂ (TPPO) ₂]	1160	152	363	141
[EuCl(BTFA) ₂ (TPPO) ₂]	1048	23	445	68
[EuCl(TTA) ₂ (TPPO) ₂]	807	11	396	2
[EuCl(DBM)(TTA)(TPPO) ₂]	775	56	331	29
[EuCl(DBM)(BTFA)(TPPO) ₂] ²¹	736	73	324	8
[EuCl(BTFA)(TTA)(TPPO) ₂] ²¹	740	47	263	84

^aIonic (either chloride or β -diketonate) $A_{\text{rad}}^{\text{ionic}}$, or nonionic $A_{\text{rad}}^{\text{nonionic}}$ coordinated to europium(III) for each of the eight chloride containing europium(III) complexes.

Table 3. Effect of a Single Ionic Coordination Symmetry Breaker on Nonradiative and Radiative Decay Rates, A_{rad} and A_{nrad} of Europium Complexes [Eu(BTFA)₃(TPPO)₂]¹³ and [Eu(TTA)₃(TPPO)₂]¹³

β -diketonate	coordination symmetry breaker	complex	$A_{\text{rad}} \text{ (s}^{-1}\text{)}$	$A_{\text{nrad}} \text{ (s}^{-1}\text{)}$
BTFA		[Eu(BTFA) ₃ (TPPO) ₂] ¹³	919	1806
	Cl	[EuCl(BTFA) ₂ (TPPO) ₂]	1142	845
	BTFA	[EuCl ₂ (BTFA)(TPPO) ₃]	1063	652
	TTA	[Eu(BTFA) ₂ (TTA)(TPPO) ₂] ¹⁹	1058	1126
	BTFA	[Eu(BTFA)(TTA) ₂ (TPPO) ₂] ¹⁹	1093	1206
	DBM	[Eu(BTFA) ₂ (DBM)(TPPO) ₂] ¹⁹	1131	983
	BTFA	[Eu(BTFA)(DBM) ₂ (TPPO) ₂] ¹⁹	1015	1349
		average	1084	1027
		[Eu(TTA) ₃ (TPPO) ₂] ¹³	796	2061
	Cl	[EuCl(TTA) ₂ (TPPO) ₂]	814	1539
TTA	TTA	[EuCl ₂ (TTA)(TPPO) ₃] ²¹	1061	1090
	BTFA	[Eu(TTA) ₂ (BTFA)(TPPO) ₂] ¹⁹	1093	1206
	TTA	[Eu(TTA)(BTFA) ₂ (TPPO) ₂] ¹⁹	1058	1126
	DBM	[Eu(TTA) ₂ (DBM)(TPPO) ₂] ¹⁹	1052	1307
	TTA	[Eu(TTA)(DBM) ₂ (TPPO) ₂] ¹⁹	1082	1328
		average	1027	1266

1, the optimized geometry of the europium complex is deemed not to be consistent with the experimental intensity parameters $\Omega_{\lambda, \text{exp}}$. From Table S1, all D/C values for all studied complexes are greater than 1. So, all $\Omega_{\lambda, \text{theo}}$ values are compatible with the corresponding experimental values $\Omega_{\lambda, \text{exp}}$. For example, the D/C ratio calculated for the [EuCl₂(TTA)(TPPO)₃] complex was 1.51.

The $\Omega_{\lambda, \text{exp}}$ with $\lambda = 2$ and 4 obtained from the emission spectrum of the complex were 32.64×10^{-20} and 6.75×10^{-20} cm², respectively, and compare with the corresponding fitted values obtained from LUMPAC,²⁹ that were 32.62×10^{-20} and 6.76×10^{-20} cm², respectively. Therefore, the theoretical model can be considered adjusted to the experimental data. It was further possible to calculate the Ω_6 value, which could not be measured from the experimental emission spectrum.

Finally, using LUMPAC software, we performed the actual calculations of the chemical partition of radiative decay rates, A_{rad}' .²⁴ For example, Figure 4 shows the effects of each of the ligands on A_{rad}' according to the chemical partition for the complexes [EuCl₂(BTFA)(TPPO)₂], [EuCl(BTFA)₂(TPPO)₃], and [EuCl(TTA)(BTFA)(TPPO)₂].

Now, we are in position to evaluate the effects of types of ligands, such as chlorides, β -diketonates, and nonionic ones (TPPO), on the radiative decay rates of luminescence of the novel europium complexes studied in this article. Table 2 shows the average effects per ligand according to the chemical partition, categorized by types of ligands on the radiative decay

rates corresponding to $^5D_0 \rightarrow ^7F_J$, where $J = 2, 4$, and 6 electronic transitions.

From Table 2 it is possible to immediately verify that the chloride ligand effects on the radiative decay rates are very small, around 60 s^{−1}, reinforcing their role as essentially symmetry breaking species, not as forthright facilitators of radiative decay.

As mentioned in the previous sections, the presence of three TPPOs, two of them opposite to each other, and the third adjacent to one of them, breaks the centrosymmetry of the complex in a significant manner. That is why the average effect of the TPPOs in the complexes of type [EuCl₂(β)(TPPO)₃] is 235 s^{−1}, which is much larger than that in complexes of the types [EuCl(β)₂(TPPO)₂] and [EuCl(β)(β')(TPPO)₂], where this value is 55 s^{−1}.

In complexes of the type [EuCl₂(β)(TPPO)₃], A_{rad}' is dominated by the contributions of the three TPPOs, each supplying 235 s^{−1}, for an average total of 705 s^{−1}. On the other hand, for complexes of the types [EuCl(β)₂(TPPO)₂] and [EuCl(β)(β')(TPPO)₂], A_{rad}' is now governed by the β -diketonate ligands, whose total contributions average 707 s^{−1}. Once again, this inversion of preponderance can be understood in geometric terms. Because the bulky TPPOs are opposite to each other, the β -diketonate ligands then end up being adjacent to each other due to the presence of the chloride ligand. Hence, the β -diketonate ligands are forced into a non-centrosymmetric arrangement, thus dominating the effect of making the radiative

transition less forbidden. The chemical partition thus clarifies the otherwise unexpected role of the ligand dissimilarity enhancer chloride.

2.7. Effect of Ligand Dissimilarity on A_{nrad} . Nonradiative decay processes tend to depopulate the excited levels of the Eu(III) complexes, most notably the $^5\text{D}_0$ state, thus reducing luminescence. Lessening the nonradiative decay rate is therefore highly desirable and a very important research goal.

Unfortunately, despite the insights provided by earlier research,^{32–34} many factors involved in determining the nonradiative decay rate still elude us.

As a matter of fact, A_{nrad} values display much wider variations among complexes than A_{rad} values. In Table 1, the standard deviation of all A_{rad} values present is 209 s^{-1} , whereas the same quantity for A_{nrad} is 1266 s^{-1} , which is 6 times larger. In addition, although the difference between the largest A_{rad} value in Table 1 from its lowest is 948 s^{-1} , the same for A_{nrad} is a whopping 5699 s^{-1} , again 6 times larger.

Therefore, detecting what affects A_{nrad} , is truly more important for luminescence than controlling what affects A_{rad} .

Our present results point in the direction that ligand dissimilarity is seemingly even more important for nonradiative decays than for radiative ones.

Take, for example, the data in Table 3, which evidence what happens to A_{rad} and A_{nrad} when going from complexes with three identical ionic ligands to complexes with two identical and a third different ionic ligand. For BTFA complexes with one coordination symmetry breaker, the average value of A_{rad} is 1084 s^{-1} , whereas the corresponding value for the complex without a coordination symmetry breaker, $[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$, A_{rad} is 919 s^{-1} , which is a difference of only $+165\text{ s}^{-1}$. That the coordination symmetry breaker species does seem to affect more intensively the nonradiative decay rates, A_{nrad} , can be verified by the fact that the average value of A_{nrad} is 1027 s^{-1} for these BTFA complexes with one coordination symmetry breaker, whereas the corresponding value of A_{nrad} without a coordination symmetry breaker, $[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$, is 1806 s^{-1} , for a difference of -779 s^{-1} . Note that the absolute value of -779 s^{-1} is almost 5 times that of $+165\text{ s}^{-1}$.

Similar figures for TTA complexes are 1027 and 796 s^{-1} for A_{rad} , for a difference of $+231\text{ s}^{-1}$. Likewise, for A_{nrad} , the figures are 1266 and 2061 s^{-1} , for a difference of -795 s^{-1} . Again, please note that the absolute value of -795 s^{-1} is more than 3 times that of $+231\text{ s}^{-1}$.

Clearly, ligand dissimilarity has a double desirable effect by both increasing A_{rad} and at the same time, by reducing A_{nrad} .

However, the effect of ligand dissimilarity on reducing A_{nrad} is much stronger.

Following this reasoning, we could presume that a fully mixed ionic ligand complex would then display even lower values of A_{nrad} . From the data in Table 4, for both the BTFA and TTA cases, A_{nrad} is indeed lowered by going from the nonmixed ionic ligand complexes to the fully mixed ones. However, the lowest A_{nrad} values occur for intermediary complexes of the type $[\text{Eu}(\text{L})_2(\text{L}')(\text{TPPO})_2]$, as is clear from Table 3, and not for the fully mixed ones of the type $[\text{Eu}(\text{L})(\text{L}')(\text{L}'')(\text{TPPO})_2]$ in Table 4. Too much diversity does not seem to be advantageous for A_{nrad} .

Now, we are faced with a choice, we can either increase ligand dissimilarity by going from the three β -diketonates to either (i) one β -diketonate and two chlorides, to (ii) two β -diketonates and one chloride, or (iii) two identical β -

Table 4. Effect of Fully Diversified Ionic Ligand Coordination on Radiative and Nonradiative Decay rates, A_{nrad} and A_{rad} , for the Europium Complexes $[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$ ¹³ and $[\text{Eu}(\text{TTA})_3(\text{TPPO})_2]$ ¹³

β -diketonate	complex	A_{rad} (s^{-1})	A_{nrad} (s^{-1})
BTFA	$[\text{Eu}(\text{BTFA})_3(\text{TPPO})_2]$ ¹³	919	1806
	$[\text{EuCl}(\text{DBM})(\text{BTFA})(\text{TPPO})_2]$ ²¹	806	1574
	$[\text{EuCl}(\text{TTA})(\text{BTFA})(\text{TPPO})_2]$ ²¹	823	1541
	$[\text{Eu}(\text{DBM})(\text{TTA})(\text{BTFA})(\text{TPPO})_2]$ ¹⁹	1034	1269
TTA	$[\text{Eu}(\text{TTA})_3(\text{TPPO})_2]$ ¹³	796	2061
	$[\text{EuCl}(\text{DBM})(\text{TTA})(\text{TPPO})_2]$	838	1707
	$[\text{EuCl}(\text{TTA})(\text{BTFA})(\text{TPPO})_2]$ ²¹	823	1541
	$[\text{Eu}(\text{DBM})(\text{TTA})(\text{BTFA})(\text{TPPO})_2]$ ¹⁹	1034	1269

diketonates and a third different one. Our present results indicate that it is much better to choose (i), that is, the situation of one single antenna with high efficiency, which we suspect might probably have more to do with the radiative processes, and two chlorides, which we suspect have more to do with the nonradiative processes, and the reason might possibly be because a chloride is much more dissimilar to a β -diketonate than another β -diketonate.

3. CONCLUSIONS

Our intention in this article was not to arrive at complexes with excellent efficiencies but rather to unveil factors and aspects affecting luminescence to further strengthen knowledge of this phenomenon on a more fundamental level.

Results point in the direction that ligand dissimilarity, ligand diversity, and centrosymmetry are all concepts that seemingly play a fundamental role in the luminescence of europium complexes. Indeed, in a previous article, we had already shown that luminescence could be boosted¹³ by moving from a homoleptic to a heteroleptic coordination.

Our results further revealed that the effects of chloride ligands on the radiative decay rates are indeed very small, around 60 s^{-1} (according to the chemical partition of A_{rad}), reinforcing their role as essentially ligand dissimilarity enhancers and not as forthright facilitators of radiative decay.

Truly, more than centrosymmetry breaking, it is ligand diversity and ligand dissimilarity that enhance luminescence, mainly by reducing the nonradiative decay rate, and the classical parity rule and its relaxation have very little to do with nonradiative decays, but the fact is that nonradiative decay rates are much more affected by ligand dissimilarity than by radiative decay rates, by an order of magnitude.

On the other hand, chloride, by being a single atom, turned out to be a very efficient coordination symmetry breaker and ligand dissimilarity enhancer, capable of boosting luminescence in a significant manner due to its unexpected strong effect on the nonradiative decay rate A_{nrad} .

Indeed, we found out that the chemical nature of the coordination symmetry breaking ligand per se is seemingly not relevant. It is important, however, that a symmetry breaker ligand exists and that ligand dissimilarity, or ligand diversity, is enhanced, all affecting mainly A_{nrad} by reducing it.

The presence of at least one antenna ligand is nevertheless obviously necessary. Therefore, the optimal and the most cost-effective situation can be achieved by adding a single symmetry breaker that is also an efficient antenna, such as TTA or BTFA. In such cases, the quantum efficiency is boosted tremendously

Table 5. Reagents and Solvents Employed in the Synthesis Procedures

reagent/solvent	source	purity (%)
1,3-diphenylpropane-1,3-dione (DBM)	Alfa Aesar	99
4,4,4-trifluoro-1-phenyl-1,3-butanedione (BTFA)	Alfa Aesar	99
1-(2-thenoyl)-3,3,3-trifluoroacetone (TTA)	Alfa Aesar	99
triphenylphosphine oxide (TPPO)	Sigma-Aldrich	98
ethanol	J.T. Baker	99.9 (high-performance liquid chromatography (HPLC))
chloroform	J.T. Baker	99.9 (HPLC)

to a maximum of $\eta = 62\%$ for the novel complex $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$. The cost effectiveness of this unprecedented strategy is due to the fact that chloride ions are much less expensive than any efficient antenna ligand.

These are easy to use chemical ideas, independent of the chemical nature of the ligands, that we devised and proved experimentally.

In summary, our results indicate that a single ligand dissimilarity enhancer antenna of high efficiency substituted in, or added to, an otherwise weakly luminescent europium complex is enough to significantly boost its luminescence.

4. EXPERIMENTAL SECTION AND COMPUTATIONAL METHODS

4.1. Materials. Table 5 shows the reagents and solvents used, their sources, and purity degrees.

The structures of both types of ligands considered in this work are shown in Figure 5.

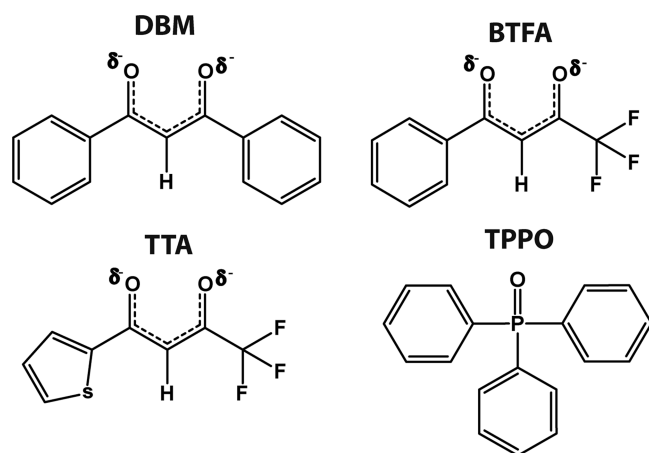


Figure 5. Structures of the ligands considered in this work: DBM, BTFA, TTA, and TPPO.

4.2. Syntheses of the Novel Chloride Europium Complexes. Previously, our research group reported a strategy for the synthesis of mixed-ligand europium complexes with chloride and β -diketonate ligands, exemplified by the synthesis of $[\text{EuCl}_2(\text{TTA})(\text{TPPO})_3]$ from $[\text{EuCl}_2(\text{TPPO})_4]\text{Cl}\cdot 3\text{H}_2\text{O}$.²¹ This same strategy was followed for the syntheses of all five novel complexes $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$, $[\text{EuCl}(\text{TTA})_2(\text{TPPO})_2]$, $[\text{EuCl}(\text{BTFA})_2(\text{TPPO})_2]$, $[\text{EuCl}(\text{DBM})_2(\text{TPPO})_2]$, and $[\text{EuCl}(\text{DBM})(\text{TTA})(\text{TPPO})_2]$, all starting from 1 mmol of $[\text{EuCl}_2(\text{TPPO})_4]\text{Cl}\cdot 3\text{H}_2\text{O}$ as well. In all cases, after the first addition of 1 equiv of a ligand, we let the mixture sit under reflux for 16 h and after the addition of another equivalent of either the same ligand or of another ligand, we let the mixture sit under stirring and reflux for 24 more hours. The differences were as follows: (i) for the

synthesis of $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$, 1 equiv of BTFA was used; (ii) for $[\text{EuCl}(\text{TTA})_2(\text{TPPO})_2]$, $[\text{EuCl}(\text{BTFA})_2(\text{TPPO})_2]$, and $[\text{EuCl}(\text{DBM})_2(\text{TPPO})_2]$, 2 equiv of the ligands, either TTA, BTFA, or DBM, were used; and (iii) for $[\text{EuCl}(\text{DBM})(\text{TTA})(\text{TPPO})_2]$, we first used 1 equiv of DBM and then 1 equiv of TTA.

4.3. Characterization. Table 6 shows the characterization techniques carried out in this article, together with the respective equipment used.

Table 6. Types of Characterization Analyses and Equipment Used

analyses	equipment
matrix-assisted laser desorption ionization time-of-flight mass spectrometry (MALDI-TOF MS)	Autoflex 3 Smart Beam Vertical spectrometer
elemental analysis	PerkinElmer CHN2400
infrared spectroscopy	Bruker model IFS 66 spectrophotometer
¹ H NMR spectroscopy	Varian Unity Plus 400 MHz
³¹ P NMR spectroscopy	Varian Unity Plus 400 MHz
¹⁹ F NMR spectroscopy	Varian Unity Plus 400 MHz

All samples for infrared spectroscopy experiments were prepared as disks of KBr. ¹H NMR, ³¹P NMR, and ¹⁹F NMR spectroscopy experiments were carried out in CDCl₃ solutions. All characterization assignments of these IR, and ¹H, ³¹P, and ¹⁹F NMR spectra are presented in the Supporting Information.

During our theoretical modeling of the structures, we could construct several possible coordination isomers for the synthesized complexes. By analyzing the lifetime curves for the complexes, we could not detect any summation of two or more different exponential decays. Therefore, from the perspective of luminescence, all solutions behave as if there is only one compound present in each one. Indeed, if there are two, or even more coordination isomers present, they all have identical luminescent properties within the resolution of our equipment. Hence, possible mixes of very similar isomers, evidenced mainly in some of the ¹⁹F and ³¹P NMR spectra present in the Supporting Information, do not invalidate our results and conclusions that focus on photophysical properties and luminescence. So far, no techniques have been developed by our group to isolate several very similar coordination isomers of a given europium complex from each other. Future research must address this issue.

4.3.1. Complex $[\text{EuCl}_2(\text{BTFA})(\text{TPPO})_3]$. **4.3.1.1. Characterization.** Calcd MALDI-TOF MS $[\text{M} + \text{H}]^+$ (m/z) 1274.16, found (m/z) 1274.31; elemental analysis calcd C 60.34%, H 4.11%, found C 59.97%, H 4.06%; IR (KBr): $\nu_{\text{C-H}}$ 3055 cm^{-1} , $\nu_{\text{C=O}}$ 1625 cm^{-1} , $\nu_{\text{P=O}}$ 1180–1185 cm^{-1} , and $\nu_{\text{C-F}}$ 1075 cm^{-1} ; ¹H NMR (400 MHz, CDCl₃): δ 8.55–6.71 ppm (m, Ar); ³¹P NMR (162 MHz, CDCl₃): δ 28 and –74 ppm;

and ^{19}F NMR (376 MHz, CDCl_3): δ –80 and –82 ppm. Yield: 84%.

4.3.2. Complex $[\text{EuCl}(\text{TFA})_2(\text{TPPO})_2]$. **4.3.2.1. Characterization.** Calcd MALDI-TOF MS $[\text{M} + \text{H}]^+$ (m/z) 1189.05, found (m/z) 1189.13; elemental analysis calcd C 52.56%, H 3.39%, found C 52.39%, H 3.30%; IR (KBr): $\nu_{\text{C-H}}$ 3055 cm^{-1} , $\nu_{\text{C=O}}$ 1680 cm^{-1} , $\nu_{\text{P=O}}$ 1180–1120 cm^{-1} , $\nu_{\text{C-F}}$ 1075 cm^{-1} , and $\nu_{\text{S=C}}$ 1070 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3): δ 9.66–7.32 ppm (m, Ar) and δ 7.01–6.27 ppm (m, Th); ^{31}P NMR (162 MHz, CDCl_3): δ 28 ppm; and ^{19}F NMR (376 MHz, CDCl_3): δ –84 and –87 ppm. Yield: 70%.

4.3.3. Complex $[\text{EuCl}(\text{BTFA})_2(\text{TPPO})_2]$. **4.3.3.1. Characterization.** Calcd MALDI-TOF MS $[\text{M} + \text{H}]^+$ (m/z) 1177.14, found (m/z) 1176.98; elemental analysis calcd C 57.18%, H 3.77%, found C 57.42%, H 3.93%; IR (KBr): $\nu_{\text{C-H}}$ 3055 cm^{-1} , $\nu_{\text{C=O}}$ 1680 cm^{-1} , $\nu_{\text{P=O}}$ 1180–1120 cm^{-1} , and $\nu_{\text{C-F}}$ 1070 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3): δ 8.55–7.33 ppm (m, Ar); ^{31}P NMR (162 MHz, CDCl_3): δ 25 and –64 ppm; and ^{19}F NMR (376 MHz, CDCl_3): δ –80 ppm. Yield: 81%.

4.3.4. Complex $[\text{EuCl}(\text{DBM})_2(\text{TPPO})_2]$. **4.3.4.1. Characterization.** Calcd MALDI-TOF MS $[\text{M} + \text{H}]^+$ (m/z) 1193.23, found (m/z) 1193.34; elemental analysis calcd C 66.47%, H 4.56%, found C 66.62%, H 4.65%; IR (KBr): $\nu_{\text{C-H}}$ 3054 cm^{-1} , $\nu_{\text{C=O}}$ 1595 cm^{-1} , and $\nu_{\text{P=O}}$ 1120–1115 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3): δ 11.98–7.35 ppm (m, Ar); and ^{31}P NMR (162 MHz, CDCl_3): δ 31 ppm. Yield: 90%.

4.3.5. Complex $[\text{EuCl}_2(\text{DBM})(\text{TFA})(\text{TPPO})_2]$. **4.3.5.1. Characterization.** Calcd MALDI-TOF MS $[\text{M} + \text{H}]^+$ (m/z) 1191.14, found (m/z) 1191.20; elemental analysis calcd C 59.53%, H 3.98%, found C 59.55%, H 4.09%; IR (KBr): $\nu_{\text{C-H}}$ 3055 cm^{-1} , $\nu_{\text{C=O}}$ 1600 cm^{-1} , $\nu_{\text{P=O}}$ 1120–1180 cm^{-1} , $\nu_{\text{C-F}}$ 1070 cm^{-1} , and $\nu_{\text{S=C}}$ 1068 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3): δ 8.15–7.32 ppm (m, Ar) and δ 7.00–6.00 ppm (m, Th); ^{31}P NMR (162 MHz, CDCl_3): δ 29 ppm; and ^{19}F NMR (376 MHz, CDCl_3): δ –75 and –79 ppm. Yield: 74%.

4.4. Luminescence. The photophysical properties emission quantum efficiency, η , radiative decay rates, A_{rad} , and non-radiative decay rates, A_{nrad} were determined from excitation spectra, emission spectra, and lifetime curves. These measurements were carried out by using 10^{-4} M chloroform solutions of the europium complexes with chloride ligands considered in this work. The maxima in the excitation spectra for all samples were found in the region from 362 to 400 nm for emissions at a wavelength of 612 nm, corresponding to the hypersensitive transition from $^5\text{D}_0$ to $^7\text{F}_2$ for the trivalent europium ion.

The slit used in the experiments was 1.0 nm for both emission and excitation spectroscopic measurements. All photophysical experiments were performed at room temperature, close to 25 °C, using a Fluorolog-3 Horiba Jobin Yvon with a Hamamatsu R928P photomultiplier, a SPEX 1934 D phosphorimeter, and a 150 W pulsed xenon lamp.

All excitation spectra, emission spectra, and lifetime curves are presented in the [Supporting Information](#).

4.5. Computational Details. The computational strategy was the one described in our previous article,¹⁹ in which we employed the semiempirical Sparkle/RM1²⁷ model, and the quantum chemical software MOPAC 2016.³⁵ Accordingly, the first step of the computational strategy was to calculate the fully optimized geometries of the chloride ligand europium complexes considered in this article. We further calculated all vibration modes for these complexes to make sure there were no imaginary frequencies and that the optimized structures were true minima. Finally, by using LUMPAC software,²⁸ we

determined the luminescence intensity parameters Ω_2 , Ω_4 , and Ω_6 , as well as the emission quantum efficiency, η , the radiative decay rate, A_{rad} , and the nonradiative decay rate, A_{nrad} , all assuming the refractive index of the medium to be 1.45, because the measurements of excitation spectra, emission spectra, and lifetime curves were performed for the chloride europium complexes in chloroform solutions.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acsomega.7b00647](https://doi.org/10.1021/acsomega.7b00647).

Luminescence data: excitation spectra, emission spectra, and lifetime curves; characterization data: infrared and ^1H , ^{19}F , and ^{31}P NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Prof. Wagner Faustino from the Department of Chemistry of the Federal University of Paraíba, Brazil for the luminescence measurements, Prof. Maria Fernanda Pimentel of the Department of Chemical Engineering of the Federal University of Pernambuco, Brazil for the infrared experiments, and Dr. Júlia F. Campos of the “Centro de Tecnologias Estratégicas do Nordeste”, CETENE for the MALDI-TOF MS data. We also acknowledge the following Brazilian agencies: “Conselho Nacional de Desenvolvimento Científico e Tecnológico”, CNPq; “Fundação de Amparo à Ciência e Tecnologia de Pernambuco”, PRONEX/FACEPE; and “Coordenação de Aperfeiçoamento de Pessoal de Nível Superior”, CAPES.

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